Shaik Atiq Rehaman

Vijayawada, Andhra Pradesh +91 8019221677
atiqrehamanshaik@gmail.com
linkedin.com/in/atiq-rehaman-shaik github.com/AtiqRehaman

Professional Summary

Computer Science undergraduate specializing in **Artificial Intelligence and Quantum Communication**. Experienced in hybrid quantum–classical systems, Superdense Coding simulation using Qiskit, and deep learning frameworks including PyTorch and TensorFlow. Passionate about bridging quantum information theory with practical AI applications.

Education

B.Tech in Computer Science and Engineering
Lakireddy Bali Reddy College of Engineering
2023 – Present

Intermediate
Sri Chaitanya Junior College
2021 – 2023

Secondary Education
Sri B V Subba Reddy Muncipal Corporation High School
2021

Technical Skills

Programming:
Python, C, Java

AI / ML:
PyTorch, TensorFlow, GANs, Diffusion Models, Computer Vision

Quantum Information:
Superdense Coding (SDC), Quantum Entanglement, Qiskit, AerSimulator, Noise Modeling, Hybrid Quantum–Classical Systems

Web:
HTML, CSS, JavaScript, MERN

Tools:
Git, Visual Studio Code, Hugging Face

Workshops

- 1-Week Workshop on Data Preprocessing, Data Analysis, and Machine Learning using Python.
- 2-Week MERN Stack Development Workshop (MongoDB, Express.js, React, Node.js).

Certifications

- Amaravati Quantum Valley Hackathon 2025 Boot Camp certificate.
- AWS Academy Graduate – Generative AI Foundations
- Google Cloud – Introduction to Generative AI
- Programming Fundamentals using Python

Research/Publications

Hybrid Quantum–Classical Image Transmission via Superdense Coding Submitted to the 8th IEEE International Conference on Devices, Circuits and Systems (ICDCS).

Projects

Hybrid Quantum–Classical Image Transmission via Superdense Coding

- Designed a hybrid communication framework combining Superdense Coding (2 classical bits per qubit) with classical transmission.
- Implemented entropy-based block selection for efficient quantum resource allocation.
- Modeled effective channel capacity using Shannon's theorem.
- Simulated circuits in Qiskit (AerSimulator with noise models).
- Achieved $\sim$16% effective transmission gain with 15% quantum allocation.

AI-Driven Crop Disease Prediction using YOLOv5

- Trained and fine-tuned YOLOv5 on custom agricultural datasets.
- Improved detection accuracy via dataset preprocessing and augmentation.
- Built inference pipeline for real-time prediction.

Bulk Image Resizer – Python CLI Tool

- Developed a command-line tool to resize single images or entire folders with custom width and height.
- Added support for PNG/JPG output formats, random sampling of N images, and automatic directory handling.
- Packaged and published the tool as a digital product with documentation and versioning.

Achievements

- Semi-Final Runner-Up – Amaravati Quantum Valley Hackathon 2025
- 2nd Place – LBRCE Coding Contest (2025)
- Consolation Prize - 8 hours MERN Hackaton (2025)

Additional Information

Languages: English, Urdu, Hindi, Telugu
Interests: Generative Models, Quantum Systems, Web Development